
Notas de charla — IIFET 2026, Tórshavn

Sesión: jueves 20 ago, 13:30–15:00 · Spatial Trade-offs of EBM (Part 2) · hablamos últimos

Paper: *Are fishing decisions flexible? Participation, species target, and landing location choices in the U.S. West Coast CPS fishery.* Quezada-Escalona, Tommasi, Kaplan, Muhling & Stohs — *Ecological Economics* **247** (2026).

Sesión: “Spatial Trade-offs of Ecosystem-based Management: Models, Theory, and Performance (Part 2)” — jueves 20 ago, 13:30–15:00, Dansistovan. **Formato real:** 7 papers en 90 min $\Rightarrow$ ~12 min por slot. Hablamos **últimos**, así que hay que asumir que la sesión va atrasada. **Objetivo:** 10 min de charla + 2 de preguntas. Plan B: 6–7 min (ver §Plan comprimido).

Cómo usar este documento: la estructura y los comentarios están en español; **las líneas entre comillas son lo que se dice en inglés**, tal cual. No hay que memorizarlas palabra por palabra — sí memorizar la *primera frase* y la *frase de transición* de cada slide, que es donde uno se traba.

Reloj de la charla

#	Slide	Tiempo	Acumulado
1	Título	0:20	0:20
2	Motivation	0:50	1:10
3	The contribution (hero)	1:30	2:40
4	Why that swap matters	0:55	3:35
5	Setting	0:40	4:15
6	The model	0:50	5:05
7	Drivers (tabla)	1:00	6:05
8	Scenario (payoff)	1:20	7:25
9	Substitution [CUT]	0:45	8:10
10	Takeaways	0:55	9:05
11	Thanks	0:15	9:20

Los dos slides que **no** se sacrifican son el **3** (la contribución) y el **8** (el payoff). Todo lo demás es negociable.

Slide 1 — Título

Idea única: quién soy y de qué va esto, en una frase.

Guion:

“Thank you. I’m Felipe Quezada, from Universidad de Concepción, and this is joint work with colleagues at NOAA. The paper asks a simple question: **when the ocean changes, how flexible are fishing decisions?** And the way we answer it is by taking a model that oceanographers built — a species distribution model — and using its daily output as the central economic regressor.”

Cuidado: no leer la lista de coautores ni los agradecimientos. 20 segundos, se pasa.

Transición: “Let me start with why the *portfolio* of species is the right place to look.”

Slide 2 — Motivation: los trade-offs espaciales pasan por el portafolio de especies

Idea única: el resultado espacial no se decide sólo en el espacio — se decide en *a qué especie te puedes cambiar*.

Guion:

“Fishable area is shrinking — offshore wind, conservation closures, mining — while the species themselves are moving with climate. For management, what actually matters is not the shock, it’s **how the fleet redistributes effort**: they can follow the species, switch species, switch ports, stay in port, or leave the fishery altogether.

The problem is that location-choice models almost always model **one fishery at a time**. When you do that, the species portfolio drops out of the model by construction — and with it, most of the adaptation margin.

So our question is: how do **availability, prices, costs and weather** shape the **daily** decision to participate, to target a species, and to land at a port? And we ask it in a period with two big natural experiments: the **2015 sardine closure** and the **2014–2016 marine heatwave**.”

Puntero: la columna derecha, las tres palabras en gold (*participate, target, land*).

Transición (memorizar): “So we model those three decisions **jointly** — and the input that makes that possible is on the next slide.”

Slide 3 — La contribución (HERO) ★

Idea única: *el modelo de los oceanógrafos se convierte en el regresor de los economistas*. Éste es **el** slide de la charla. Si sólo se entiende uno, que sea éste. Ir más lento aquí que en cualquier otro.

Guion:

“Here is what we do differently. Oceanographers build species distribution models — SDMs — for **ecological** purposes: they take the ocean state from a ROMS model, sea surface temperature, sea surface height, chlorophyll, and fit a GAM that predicts the **probability of presence** of a species on a spatial grid, day by day. That’s Muhling and co-authors, 2019 and 2020.

[pausa, seguir la flecha] We take that daily field and average it **within a species-specific radius of each port**. That gives us **Availability**: an **exogenous, gap-free** measure of how much of each species is around **each port, for each species, on each day**. And that variable enters the fisher’s utility function directly.

One sentence on the assumption: we’re assuming **habitat suitability tracks potential catch rates**. If it only does so partly, our availability effect is **attenuated** — biased toward zero, not inflated.”

Frase para decir literal, mirando a la audiencia (no a la pantalla):

“In one line: **the oceanographers’ model becomes the economists’ regressor.**”

Puntero: recorrer el pipeline izquierda→derecha, arriba→abajo (Ocean state → GAM SDM → daily grid → port radius), y terminar en la caja dorada.

Cuidado:

- El mapa de la derecha **no es output nuestro** — es ilustrativo, de Chasco et al. (2022). Decirlo si alguien pregunta: *“that map is illustrative, it’s not our own SDM output — it’s there to show what a habitat field looks like.”*
- **No** decir que el SDM predice *mejor* la captura que el catch pasado. No hicimos esa comparación (ver §Q&A #1).

Transición: “Why is that swap worth doing? Three reasons.”

Slide 4 — Por qué importa el cambio: catch pasado vs. SDM

Idea única: el catch pasado es endógeno, sesgado y con huecos; el SDM no — y sobre todo, el SDM se puede *mover*.

Guion: leer sólo las palabras en negrita, no las viñetas completas.

“The standard proxy for availability is **past catch** — CPUE from previous trips or previous years. Three problems. It’s **endogenous**: it records where vessels already *chose* to fish, and that choice is exactly what we’re modeling. It’s **selection-biased**: you only observe it at sites that were fished — Chen and co-authors make this point in 2023. And it has **gaps**: it’s simply undefined for a port-day when nobody went out.

The SDM index has none of those. It’s an **environmental input**, so it’s exogenous to the vessel’s decision. It has a **value for every port-day**, including the days nobody fished. And — this is the one that buys us the second half of the talk — it’s **counterfactual-ready**: you can move the habitat field and re-predict.”

Frase de cierre del slide (importante, engancha con el payoff):

“**You cannot shift a ‘past catch’ field.** That’s the whole reason we can run a spatial scenario at all.”

Transición: “Quick word on the setting, and then the results.”

Slide 5 — Setting: cuatro segmentos de flota, cuatro portafolios

Idea única: la unidad de adaptación no es “la pesquería”, es el **segmento de flota** — cada uno con un portafolio distinto.

Guion: rápido, 40 s. No leer la tabla.

“This is the U.S. West Coast Coastal Pelagic Species fishery — market squid, sardine, anchovy, mackerels, plus non-CPS options like Dungeness crab, salmon and tuna. Purse seine.

We don't treat it as one fleet. From a cluster analysis in earlier work we get **four segments**: a large southern **squid specialist** — that's the bulk of the data, about 29,000 trips; a **roving squid-sardine generalist**; a **Pacific Northwest sardine specialist**; and a small southern **forage-fish diverse** segment. Four segments, four different portfolios. Same shock, different room to manoeuvre."

Puntero: sólo la columna "Main target(s)".

Números por si preguntan: ~41,000 trips en total (29,160 / 6,806 / 2,581 / 2,481); 2013–2017; los segmentos vienen de Quezada et al. (2024).

Transición: "The model is a standard nested logit — one slide, no derivations."

Slide 6 — El modelo

Idea única: una sola línea de utilidad; lo único que hay que retener es que **Avail entra diario y es exógena**.

Guion: (comprimible a 25 s si vamos tarde)

"Each alternative is a **triple**: participate, target species, landing port. Utility is linear: an alternative-specific constant, our **availability term** — coefficient allowed to vary by species — expected price, plus distances, wind, closures and local unemployment. And a **state-dependence** term: whether the vessel made that same choice in the last thirty days. That picks up processor ties and port capacity — the fact that these vessels have contracts, not just preferences.

We nest first on **participation**, then on **species** — except for the forage-diverse segment, where the data prefer nesting by **port**. That difference matters later. Crab and salmon are split off, because switching gear there is costly. Estimated with Apollo in R; the dominant species defines the target, and that's 93.6 % of trip revenue, so it's not a coarse assignment."

Cuidado: no derivar nada, no explicar la fórmula del nested logit, no mencionar GEV. Si alguien quiere, está en el backup de λ .

Transición: "So: what actually drives the choice?"

Slide 7 — Drivers: coeficientes de disponibilidad

Idea única: la señal del SDM sobrevive a una inercia muy fuerte $\Rightarrow$ la flexibilidad que medimos es real.

Guion:

"Rows are the availability of each species; columns are the four fleet segments. Read **signs and significance**, not magnitudes — these are utility units, so they're not comparable across segments; magnitude comes from the scenario in two slides.

Squid availability is strongly positive wherever squid is relevant. Anchovy is positive for three segments. Non-CPS availability — crab and salmon — is large and positive for the two segments that have those options: those species **gate participation**.

Now look at the **bottom two rows**, because that's the whole argument. **Prices** are positive and significant for three of four segments. And **state dependence is big** — between 2.3 and 3.1 everywhere. These fleets are **habitual**: contracts, processors, capacity."

Frase clave (decirla mirando a la audiencia):

"And yet, *even with that much inertia in the model*, the **SDM signal survives**. That's what tells us the flexibility we measure is genuine behaviour, not just persistence."

Puntero: las **dos filas de abajo**. No pasear el puntero por toda la tabla.

Si preguntan por el anchovy negativo del forage-diverse (-1.88): está en la nota al pie — la disponibilidad de anchoveta y de calamar están **correlacionadas negativamente** ($r = -0.29$), así que ese coeficiente recoge el costo de oportunidad, no aversión a la anchoveta.

Transición (memorizar): "Signs are one thing. Here's what it's worth in space."

Slide 8 — El payoff: mover el campo de hábitat ★

Idea única: movemos el hábitat del calamar y el modelo re-predice **dónde termina el esfuerzo** — y el total por puerto **esconde una recomposición de especies**.

Guion:

"This is what the SDM input buys you. We take the squid habitat field and swap it for a **later window within the same season** — the habitat centroid moves south, from about **43 degrees North to 40.7**. Everything else is held fixed: prices, costs, weather, the other species. Then we let the model re-predict.

Each panel is a fleet segment; bars are the change in the **probability of landing at each port**, in percentage points. Gold gains effort, blue loses it.

Two things. First, **Santa Barbara absorbs the squid** — plus 5.8 points for the squid specialist, plus 3.4 for the roving generalist, plus 2.1 for the forage-diverse fleet. And the fleets go out **more often**: non-participation falls by 6 and 4 points for the two southern squid fleets. The Pacific Northwest sardine fleet barely moves — it has no squid to follow.

Second, and this is the interesting one — [apuntar al panel **abajo a la derecha**, *S. CCS forage diverse*] — **Los Angeles goes down**, minus 0.6. But its *squid* landings go **up**. It loses effort because it loses **chub mackerel**, almost two full points. The port-level total **hides a recomposition of the species mix**."

Frase para cerrar (es la que habla directamente a la sesión):

"And note what this machinery is: we moved a habitat field and got a spatial reallocation. **A closed area is the same operation** — you mask the field inside the polygon and re-predict. That's the door the SDM opens."

Puntero: primero el panel superior izquierdo (squid specialist, la barra grande de Santa Barbara), después el **inferior derecho** (forage diverse) para la historia de LA. **Ojo: la historia de Los Angeles es la del panel forage-diverse** (LA = -0.59, chub mackerel = -1.88), no la del squid specialist.

Cuidado: el escenario es una **redistribución dentro de la misma temporada**, no una proyección climática. Si se dice “climate scenario”, alguien va a preguntar por el forzante y no lo hay.

Transición: “Why did LA lose effort while its squid went up? One slide.”

Slide 9 — Estructura de sustitución [CUT]

Este es el slide que se bota primero si la sesión va atrasada. *Explica* el anterior, no agrega un resultado nuevo, y el slide de takeaways igual cierra el argumento. Si se bota, decir en su lugar una sola frase al pasar al slide 10: *“and the reason LA behaves that way is that for that fleet, the closest substitute is another species, not another port.”*

Idea única: dos formas de adaptarse — *on the move* (seguir la especie entre puertos) y *in place* (cambiar de especie en el mismo puerto).

Guion:

“The nesting parameters tell you *how* each fleet adapts. Lower lambda means alternatives inside the nest are **closer substitutes**.

For most segments the nests are **species**, and the lambdas are low — anchovy 0.49, sardine 0.40. That means: given the species, the **ports** are close substitutes. Those fleets adapt **on the move** — they follow the species along the coast.

The forage-diverse segment nests by **port**: Santa Barbara 0.49 against 0.73 for Los Angeles and Monterey. There, substitution happens **across species within a port**. That fleet adapts **in place** — and that’s why Los Angeles lost effort even as its squid went up: its closest substitute was another species, and that species left.

The terminology — ‘on the move’ versus ‘in place’ — is from Samhuri and co-authors, 2024.”

Transición: “Three things to take away.”

Slide 10 — Takeaways

Idea única: SDMs sirven como regresores económicos; los pescadores son flexibles; y la respuesta espacial pasa por el portafolio.

Guion:

“Three points.

One: oceanographic SDMs work as **economic regressors** — exogenous, gap-free, daily. And the estimated models predict a **held-out year** well: out-of-sample pseudo-R-squared between **0.22 and 0.44** against the null model.

Two: harvesters are **flexible**. Participation, target and landing all respond to availability, prices, costs and weather — and they do so *even after* controlling for very strong state dependence.

Three: the spatial answer runs **through the portfolio**. Even the fleet that adapts *in place* — substituting **species** rather than ports — ends up **redistributing effort in space**. If you model one species at a time, you miss that channel entirely.

And because availability here is an **ocean-model output**, the same machinery runs forward under **climate projections**, or under a closed area. Thank you.”

✗ **Cuidado crítico:** el 0.22–0.44 es **contra el modelo nulo**, *no* contra una especificación con catch pasado. **Nunca** decir “the SDM predicts better than past catch” — no corrimos esa comparación. El argumento del SDM es de **identificación y de escenarios**, no de superioridad predictiva. (Ver Q&A #1.)

Slide 11 — Thanks

Dejarlo en pantalla durante las preguntas. No leerlo. Sólo: “*Thank you — happy to take questions.*”

Plan comprimido (6–7 min)

Hablamos últimos: es probable que haya que hacer esto. Orden de sacrificio:

1. **Botar el slide 9** (substitution) — pasar directo de 8 a 10 con la frase puente de una línea.
2. **Comprimir el slide 6** (modelo) a 25 s: “*Nested logit over participate × target × port. Availability enters daily and is exogenous; we also control for prices, distances, wind and a thirty-day state-dependence term. Estimated in Apollo.*” y pasar.
3. **Comprimir el slide 5** (setting) a 20 s: “*Four fleet segments from a cluster analysis, four different species portfolios.*”
4. Si aún así falta tiempo: **botar el 4** y meter su idea en una frase dentro del 3 (“*the usual proxy is past catch, which is endogenous, gappy, and — crucially — impossible to shift in a counterfactual*”).

Nunca botar: 3 (contribución) y 8 (payoff). Con esos dos y el 10 la charla se sostiene en 4 minutos.

Números para tener en la cabeza

Dato	Valor
Período	2013–2017 (cierre de sardina 2015; ola de calor 2014–16)
Trips totales	~41,000 (29,160 / 6,806 / 2,581 / 2,481)
Revenue de la especie dominante	93.6 %
State dependence (30 d)	+2.3 a +3.1, *** en los cuatro segmentos
Radio del SDM	60 km (sardina/anchoveta/caballa), 90 km (calamar)
Escenario: centroide del calamar	42.99°N → 40.66°N (mismo período estacional)
Δ Santa Barbara	+5.8 / +3.4 / — / +2.1 pp
Δ Los Angeles (forage diverse)	–0.6 neto (calamar +1.5, chub mackerel –1.9)
Δ participación	+6.0 / +4.2 / –0.6 / +1.9 pp
Δ target calamar	+8.4 (squid spec.) / +4.5 (roving) / +5.0 (forage div.) pp

Dato	Valor
λ más bajos	anchoveta 0.49, sardina 0.40, Santa Barbara 0.49
OOS pseudo- ρ^2 (año retenido, vs. nulo)	0.22–0.44
Correlación anchoveta–calamar (disponibilidad)	$r = -0.29$

Banco de preguntas

Formato de respuesta: **una frase de respuesta directa, una de sustento, parar**. No dar mini-charlas.

1. “¿Compararon el SDM contra el catch pasado? ¿Predice mejor?” — *La pregunta más probable, y la más fácil de contestar mal.*

“No — we didn’t run that horse race, and I want to be precise about it. Our case for the SDM is **identification and counterfactuals**, not predictive superiority: past catch is endogenous to the choice we’re modeling, and you can’t shift it in a scenario. What we *do* show is that the estimated models predict a held-out year well against the null, and Table S1 compares SDM **timing** variants — daily, one-day lag, seven-, fourteen- and thirty-day moving averages. A direct SDM-versus-past-catch comparison is a good idea for future work.”

2. “Habitat suitability no es lo mismo que tasa de captura.”

“Agreed, and that’s our main caveat. Suitability is a proxy for potential catch rates. The good news is the direction of the bias: classical measurement error in a regressor **attenuates** the coefficient. So our availability effects are, if anything, a lower bound.”

3. “¿Los precios no son endógenos?”

“Partly, yes. We use **predicted** prices from a year-trend plus month and port fixed effects, and check robustness with thirty-day moving averages. That reduces simultaneity but doesn’t eliminate it, so we read price coefficients as **reduced-form** behavioural responses, not structural supply elasticities.”

4. “¿Por qué la anchoveta sale negativa para el forage-diverse?”

“Because it’s an opportunity cost, not a dislike of anchovy. In that region anchovy and squid availability are **negatively correlated** — r of about -0.29 — so high anchovy days are low squid days for a fleet whose best option is squid.”

5. “¿Podrían correr un cierre por eólica offshore / un AMP?”

“Yes — it’s the same operation as the scenario I showed. You mask the habitat field inside the closure polygon, recompute the port-radius averages, and re-predict. We haven’t run it in this paper, but nothing in the machinery has to change. That’s exactly why the exogenous field matters.”

6. “¿Por qué las estructuras de nidos difieren entre segmentos?”

“It’s empirical. We tried both structures for each segment and kept the one the data supported — fit, and dissimilarity parameters inside the unit interval. For the forage-diverse fleet the port nesting wins, and that’s substantively informative: it is the finding about adapting in place.”

7. “¿Agregación a nivel de puerto no oculta heterogeneidad espacial fina?”

“It does — we list it as our first limitation, following Dépalle and co-authors, 2021. Our choice set is port × species, so within-port location choice is averaged out. The trade-off buys us the joint participation–target–location model; a finer grid would make the choice set intractable.”

8. “¿Cómo separan state dependence de heterogeneidad no observada?”

“Honestly, not perfectly — that’s the classic initial-conditions problem. What I’d emphasise is the direction: the thirty-day term absorbs a lot of persistence, so it makes it **harder** for availability to show up. The availability effects survive that, which is the point of the slide.”

9. “¿Por qué sólo 2013–2017?”

“Two reasons: it’s the window where the daily SDM fields and the trip-level PacFIN data overlap cleanly, and it contains the two shocks we care about — the 2015 sardine closure and the 2014–16 marine heatwave.”

10. “¿Modelan entrada y salida de embarcaciones?”

“No. Our extensive margin is the **daily** decision to go fishing or not, conditional on being in the fleet. Long-run exit and vessel investment are a different model.”

11. “¿Propagan la incertidumbre del SDM?”

“Not in this paper — availability enters as a fixed regressor. That’s measurement error, so again it attenuates. Propagating the SDM’s predictive distribution through the choice model is a clear next step.”

12. “Los coeficientes de no-CPS son enormes (+10.3 crab en PNW).”

“They are, and they’re doing a specific job: crab and salmon availability essentially **gate participation** for the fleets that have those permits. They sit in a separate nest because switching gear is costly, so those magnitudes aren’t comparable to the CPS terms.”

13. “¿Y bajo proyecciones climáticas?”

“The SDMs are driven by ROMS output, so in principle you force them with downscaled projections and run the same behavioural model forward. That’s the direction we’re taking it — with the caveat that the behavioural parameters are assumed stable, which is a strong assumption over decades.”

Backups: mapa rápido

Saber en qué orden están, para llegar de una:

1. **Descriptives** — persistencia + switching (squid specialist: 19.8 cambios de especie, run mediano 2 días). *Úsalo si preguntan por qué hay state dependence.*
2. **Elasticities** — precio mueve la **participación**, disponibilidad mueve el **targeting**. *Úsalo si preguntan por magnitudes (el slide 7 no las da).*
3. **Scenario** — **participación** (gráfico de barras). *Si preguntan cuánto más salen a pescar.*
4. **Scenario** — **por especie objetivo**. *Si preguntan qué especie ganó/perdió.*
5. **Scenario** — **por puerto × especie**. *La mejor para la historia de LA.*
6. **λ (dissimilarity)** — tabla completa. *Si cuestionan la estructura de nidos.*
7. **Variable construction** — SDM, precios, costos, viento. *Si preguntan por los radios o ERA5.*
8. **Limitations** — las cuatro, escritas. *Si la pregunta es incómoda, ir directo aquí: se ve mejor tener la limitación en un slide que improvisarla.*

Recordatorios de entrega

- **Hablamos últimos.** Preguntar al chair al llegar cuánto tiempo real hay. Si dice “eight minutes”, aplicar el plan comprimido **desde el principio**, no improvisar recortes a mitad de charla.
- Mirar a la audiencia en las cuatro frases marcadas (slides 3, 4, 7, 8). El resto se puede leer del apuntador.
- El mapa del slide 3 es **ilustrativo** (Chasco et al. 2022) — decirlo antes de que lo pregunten.
- Nunca afirmar que el SDM **predice mejor** que el catch pasado.
- Si se acaba el tiempo en el slide 8: saltar directo al 10 y decir sólo el punto 3 + la frase de cierre.